

Improving relocalization accuracy via environment prior

Lars Hecker¹, Raman Besenfelder¹, David Farah¹, Fatemeh Sadat Daneshmand², Elisabetta Fedele¹, Pan Linfei¹

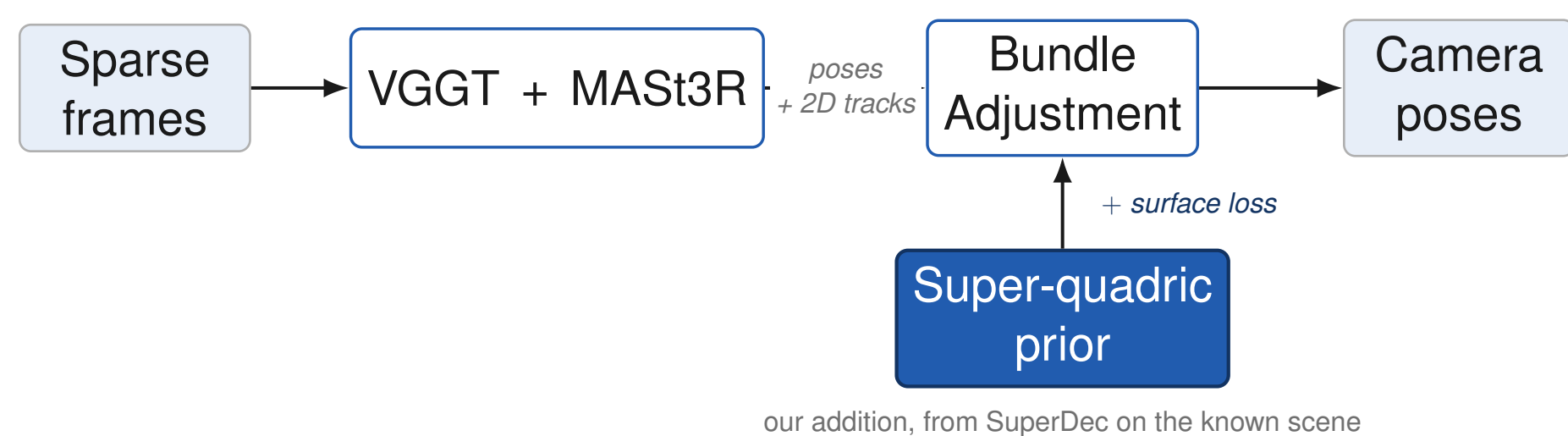
¹ETH Zurich ²ZHAW

A super-quadric environment prior, added on top of bundle adjustment, improves camera-pose accuracy. The gain is largest when views are sparse and barely overlap.
+1.5 AUC@5 at 6 views (up to +2.0 at 4). Only +0.2 once 10 views already pin the geometry.

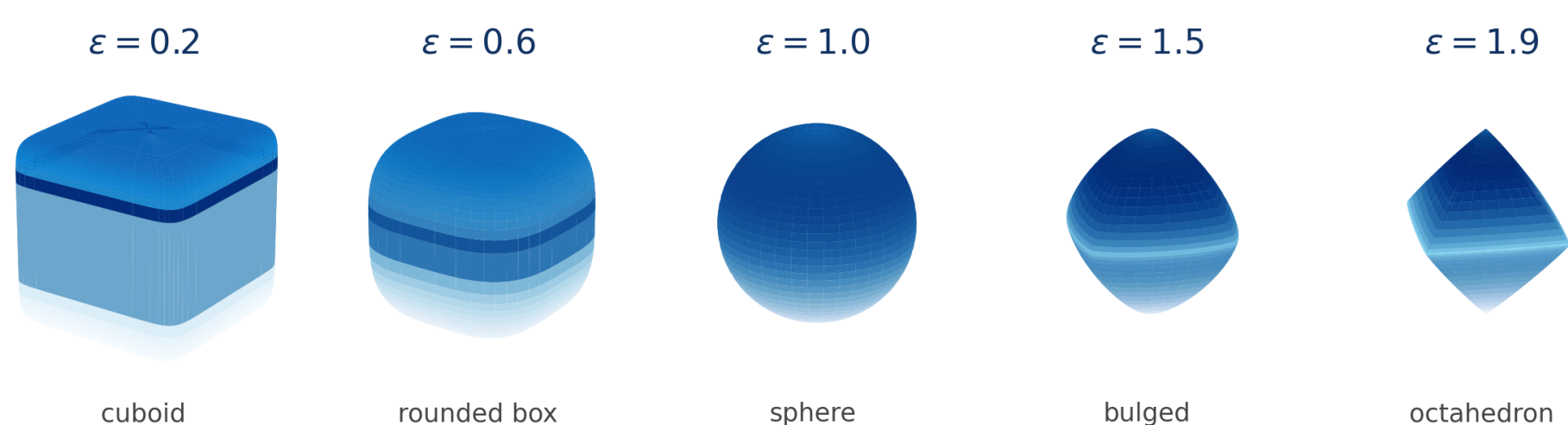
1 The Problem

Relocalization (the “kidnapped robot”) asks for an accurate camera pose in a *known* environment. Transformer-based SfM such as VGGT [1] returns fast feed-forward poses, but **cannot use such a prior**, and degrades when views are few and barely overlap. We add a **super-quadric** scene prior [4].

2 Method Overview



3 The Environment Prior



A super-quadric is one shape family. Two exponents set its roundness, three half-extents set its size, and a pose places it, so a handful tile a room.

SuperDec [4] fits a compact set of **super-quadrics** to the scene point cloud (right). During BA, a one-sided point-to-quadric term pulls each triangulated point toward its nearest primitive, with EM-style re-association every few iterations:

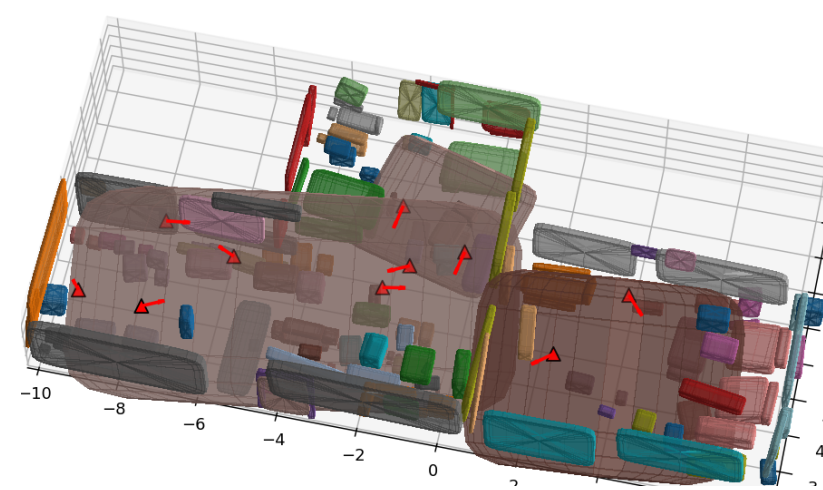


Fig. 1: a scene as 189 super-quadrics + camera poses.

$$E = \underbrace{\sum_{i,j} \rho(\|\pi(R_i, t_i, X_j) - x_{ij}\|^2)}_{\text{reprojection}} + \underbrace{\sum_j \left(\lambda \|q_j\| \cdot \max(0, 1 - F(q_j)^{-\epsilon_1}) \right)^2}_{\text{one-sided surface prior}}$$

where $F(\cdot)$ is the inside-outside function of the super-quadric nearest to point X_j and λ the prior weight [5].

4 Setup

- **Dataset:** Aria Synthetic Environments (10 scenes)
- **Models:** VGGT for feed-forward poses (via MapAnything [2]), MAST3R for matching [3]; SuperDec [4] for the prior
- **Optimization:** full triangulation + surface BA in Ceres; **metric:** pose AUC@5° (higher is better)

5 Results

Three stages: raw VGGT poses, **bundle adjustment**, then BA + **our super-quadric prior**. BA does most of the work (raw VGGT averages only 9.3 AUC@5 at 10 views, 20.4 at 6). The prior then adds a gain *on top of BA* that grows as views get sparser.

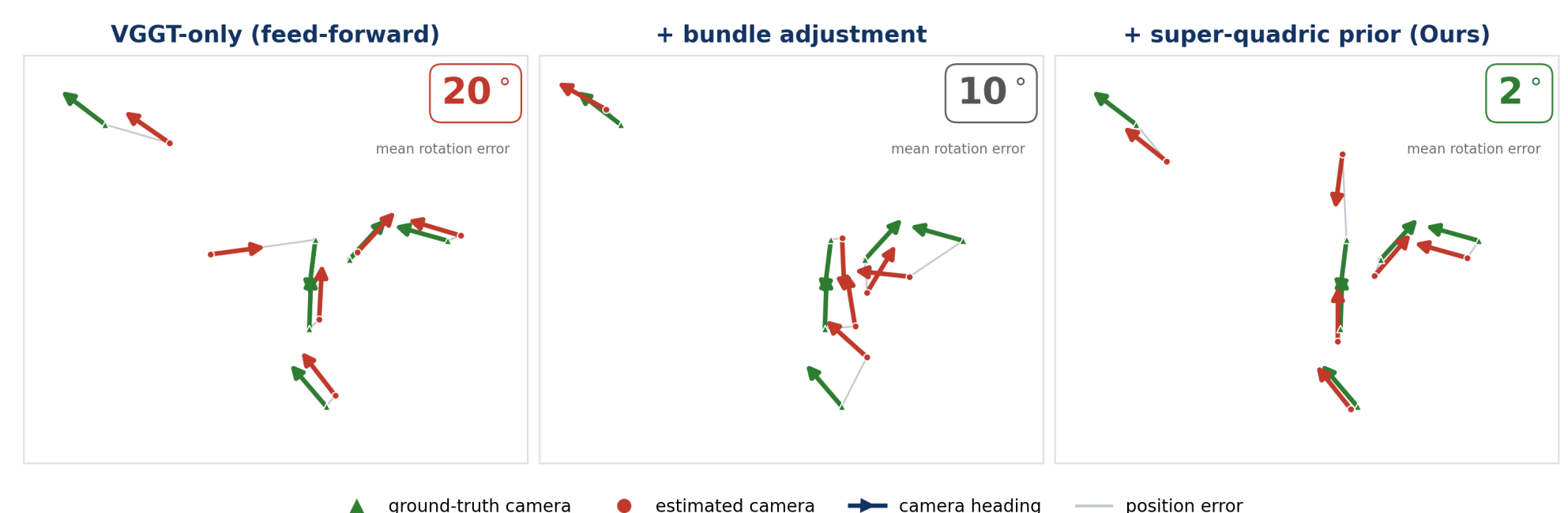
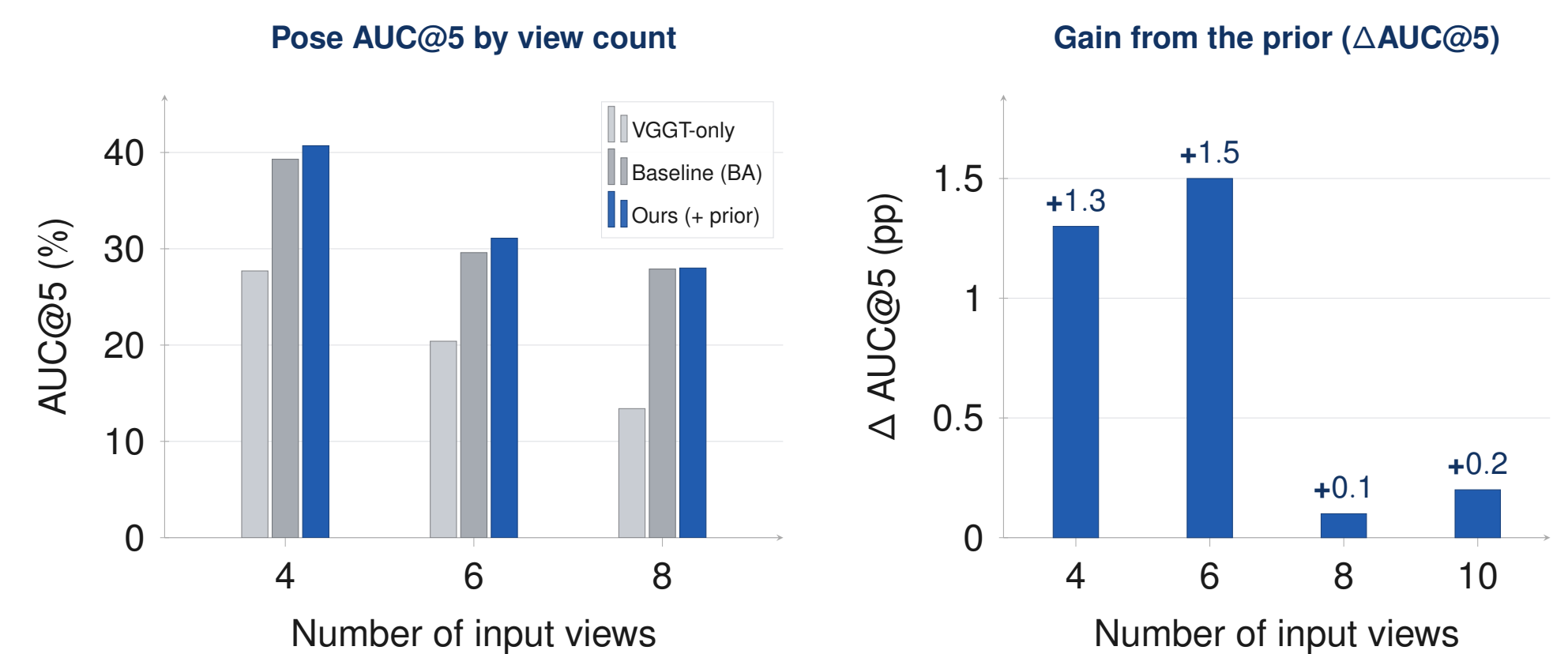


Fig. 2: one scene (scene 6, 6 views), cameras Sim(3)-aligned to ground truth for display. Estimated headings (red) move toward ground truth (green) as bundle adjustment, then the prior, are added. Mean rotation error 20° → 10° → 2° (pose AUC@5 29 → 48 → 57).



Left: bundle adjustment (grey) lifts raw VGGT (light), and our prior (blue) adds more as views get sparser. Right: the prior's gain grows from +0.2 at 10 views to +1.5 at 6 and +1.3 at 4 (up to +2.0 with light λ tuning); at 6 views it helps 3/10 scenes (up to +9.3) and hurts none.

6 Conclusion

At high view counts, dense correspondences already constrain the poses, so the prior adds little (+0.2 at 10 views). As views drop and co-visibility collapses, reprojection alone under-constrains the geometry. The super-quadric surface supplies the missing constraint, recovering up to +2.0 AUC@5 at 4 views. **An environment prior encoded as a super-quadric cloud helps relocalization most under sparse, low-covisibility views.**

References & Code

- [1] Wang et al., VGGT: Visual Geometry Grounded Transformer. CVPR 2025.
- [2] Keetha et al., MapAnything: Universal Feed-Forward Metric 3D Reconstruction. 3DV 2026.
- [3] Leroy et al., Grounding Image Matching in 3D with MAST3R. ECCV 2024.
- [4] Fedele et al., SuperDec: 3D Scene Decomposition with Superquadric Primitives. ICCV 2025.
- [5] Mueller et al., Reconstructing People, Places, and Cameras. CVPR 2025.

